DESIGN AND IMPLEMENTATION OF A VIRTUAL LAB PLATFORM FOR ACADEMIC TEACHING: EXPERIENCE IN THE DEVELOPMENT AND USE OF A NEW PLATFORM FOR THE DIGITIZATION OF LABORATORIES FOR COMPUTER SCIENCE CLASSES

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ABSTRACT

This paper presents a virtual platform for university teaching that simplifies the creation and use of complex IT setups, even for students with little to no experience. Setting up infrastructures for courses like computer networks or IT security has traditionally been challenging. However, the platform presented in this paper simplifies the setup by using virtualization techniques for compute, network, and storage resources, reducing the efforts in preparation for lecturers, researchers, and students. The platform gives access to complex IT setups utilizing free software components. The project employs Proxmox Virtual Environment and CEPH for virtualization and distributed storage, respectively, providing a flexible and practical solution. On top of that, a novel graphical web interface for the easy access and use of the virtual infrastructures is implemented using ReactFlow.

KEYWORDS

Virtual Teaching, Proxmox VE, CEPH, Teaching Computer Science, Asynchronous Teaching, Digitization of Higher Education

1. INTRODUCTION

The COVID-19 pandemic has driven the importance of digitalization into the spotlight, particularly in sectors like administration, healthcare, and education. The need for high-quality, flexible teaching solutions that can be accessed anytime, anywhere has never been more urgent. This situation presents a unique opportunity for the integration of innovative face-to-face courses with digital elements. However, the success of these initiatives is critically dependent on the availability of user-friendly IT infrastructures that can adapt to fluctuating resource demands.

Digitalization enhances university teaching with modern methods like online tests and media content, supplementing traditional concepts like laboratory practicals (Erpenbeck and Sauter, 2017). Various approaches exist, including digitized teaching concepts and online course programs (Beckmann, 2020). Crucial factors include course suitability, content selection, and provision of digitized elements (Handke, 2015).

Digital tools like Moodle (Athaya et al., 2021) differ from virtual courses. While tools like Moodle focus on specific aspects like participant registration and exercise organization, the SKILL/VL (Strategic Competence Platform - Innovative Learning and Teaching/Virtualization of distributed environments for teaching) project focuses on designing virtual teaching offerings, expanding teaching beyond face-to-face constraints through technology-based learning (Handke and Schäfer, 2012). Current teaching solutions offer limited options for designing and using complex IT infrastructures. This gap presents opportunities for innovation in teaching and research, particularly in areas like computer networks, distributed systems, IoT, Industry 4.0, and IT security. The project discussed here addresses this gap by enabling intuitive configuration and use of complex IT scenarios through virtualization, even for users with limited prior knowledge.

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The following section outlines the infrastructure's technical structure, practical use, and design decisions, covering aspects like functionality, performance, scalability, security, authorization, and authentication. It provides examples of the platform's use in teaching and then summarizes the findings and future directions for the project.

2. RELATED WORK

Projects in the field of digitized infrastructures for academia are not new, and a few examples already exist. One example is the Fed4Fire+project (Nussbaum, 2019). It consisted of large testbeds, providing open, accessible, and reliable facilities to various research and innovation communities and initiatives in Europe. The extensive setup aimed to develop and share resources for the further investigation of future networking and cloud-to-edge computing technologies. However, the project's aim was purely research-based and did not aim at establishing digital facilities for teaching.

Another example from that field is the project SLICES-RI (Demchenko et al., 2023, pp. 1-6). It was constructed to develop a large testbed for future networking, cloud, and edge computing technologies. The project aims to design and operate a complex and continuously evolving digital infrastructure for research purposes. It provides advanced computing, storage, and network components for european researchers and is intended to give researchers the resources for large-scale experimentation in developing future networking technologies. However, these resources are reserved for research purposes only and neglect the needs of scholarly activities found in institutions in the realm of higher education.

For the education in networking courses, several tools for the simulation of networks already exist on the market. An interesting tool that offers many features for the simulation of computer networks is the tool Packet Tracer, developed and distributed by Cisco (Abdul Rashid et al., 2019). Packet Tracer offers a graphical interface that is intuitive and easy to use for user groups like students, allowing the creation of simulated network topologies by offering drag-and-drop functions to configure routers, switches, and various other network devices. However, users must have a Cisco account and sign in to the Cisco Network Academy. The application only has a few features within the actual hardware and is strictly bound to Cisco networking hardware and its proprietary IOS operating software for switches and routers.

Another example is the tool mininet (Lantz et al., 2010). Mininet can set up, configure, and simulate software-defined networks. It is open-source and uses native Linux-Kernel functionalities to access process-based virtualization and network namespaces to set up virtual networks. Mininet uses software-based switches like Open vSwitch and virtual Ethernet connections for the communication between the virtual processes. However, mininet is operated over the Linux terminal and, therefore, lacks a user-friendly interface for students, which makes the learning curve very steep. Also, the components used for the interconnects, like Ethernet adapters and switches, are emulated.

GNS3 (short for Graphical Network Simulator-3) is an open-source network simulator (Emiliano et al., 2015). It allows the combination of virtual and real devices to simulate complex networks. Many large companies, including AT&T and NASA, use GNS3. GNS3 emulates a device's hardware and can run actual images on the virtual device. For example, Cisco IOS can be installed from an actual, physical Cisco router and run on a virtual, emulated Cisco router inside GNS3. GNS3 also offers simulation capabilities. It simulates features and functionalities of devices, such as a switch. These devices do not need a running operating system (such as Cisco IOS). They can run on a simulated device developed by GNS3, e.g., a built-in layer-2 switch. However, it is limited to networking scenarios and is not extensible to other scenarios. Therefore, the SKILL/VL environment focuses on extensibility and applicability to real-life scenarios.

A very interesting project in the field of digitization in higher education is presented in the work by Dietz (Dietz, 2023). He investigated the constraints and benefits of developing a virtual lab for cyber security use cases using the two public cloud service providers (CSP) Azure Virtual Desktop (AVD) and IONOS Data Center Designer (DCD). The setup with IONOS DCD seemed promising because of the easy-to-use graphical web interface, allowing students to reduce the abstraction level in the construction of complex IT scenarios. However, the work of Dietz is using the resources of a public CSP and yields high operational costs and low control of the used components. Nevertheless, IONOS DCD shows potential and, therefore, sets the basis for the implementation of the web user interface of the SKILL/VL platform. In contrast to Dietz's work, the SKILL/VL platform uses only open-source components and is implemented on-premise, giving complete control of the infrastructure.

3. DESIGNING A VIRTUAL LAB

The decision to design a virtual lab to simulate networking scenarios stems from the necessity of extending the physical lab for computer networks since the physical lab only has space for a maximum number of 24 students in a class. Another reason for designing such a virtual lab stems from the experiences during the COVID-19 pandemic. Asynchronous teaching methods using digital platforms were a pivotal element for teaching in higher education during that period and essential for successfully implementing courses without the need for face-to-face teaching. Therefore, the focus was to implement a virtual lab that is flexible and powerful enough for the operation networking courses. Table 1 presents characteristics demonstrating the benefits of using a virtual lab in higher education.

Table 1. Characteristics of physical and virtual lab (Alam, A., et al., 2023)

Characteristics	Physical Lab	Virtual Lab
Realism	High	High
Content	Stable	Dynamic
Focus of study	Lecturer	Student
Form	Synchronous	Asynchronous
Number of students	Limited	Without (physical) limits
Time	Scheduled	Anytime
Focus of course	Group	Individual
Accessibility	Low	High
Cost	Very high	Low
Maintenance Effort	Very high	Low
Remote work	Not possible	Possible

At the beginning of the project, a selection of open-source tools for the simulation of networking courses was inspected. Section 2 already presented three alternatives for the simulation of scenarios in computer networks. Table 2 presents the criteria for the different solutions that were inspected.

With the exception of Cisco, all the solutions we inspected are open-source and free of use. This criterion was pivotal, as licensing fees can significantly increase the operation costs and restrict the freedom of use. However, GNS3, the only solution that offers a graphical user interface, is another crucial criterion. This is particularly important as the tool should be accessible to students with little experience in the field of networking technologies, thereby reducing the barrier to entry. GNS3 emerged as a promising tool for the computer networks course, but it has a limitation that it can only be used in computer networks classes.

Table 2. Comparison of SKILL/VL to exisitng solutions

Criteria	SKILL/VL	Cisco Packet Tracer	Mininet	GNS3
Open Source	Yes	No	Yes	Yes
Simulator	Yes	Yes	No	Yes
Emulator	No	No	Yes	Yes
Scalability	Yes (by extending Web service)	No	limited (using more processes)	No
Automatic setup	Yes	No	No	No
Graphical User Interface	Yes	Yes	No	Yes
Extensible beyond Networks	Yes	No	No	No

The SKILL/VL platform should be extensible beyond the computer networks' application field. It should also offer capabilities for use in classes in distributed systems or software engineering, for example.

Section 4 describes the infrastructure constructed for the operation of the platform. In this section, the selection of open-source components and the design of the platform are presented with a focus on reliability. Section 5 presents the implementation of the used components and the details of the internal communication of these components.

4. INFRASTRUCTURE SKILL/VL

The virtualization platform developed in the project operates on an infrastructure consisting of 10 high-performance servers that are set up in a so-called cluster network. When designing and developing the overall system, the main objective is to develop a platform for virtual teaching in all disciplines and, in the long term, for the entire university. This long-term goal places high demands on the overall system's usability, scalability, and security.

The software stack used for the developed platform is based exclusively on free software components to create a future-proof and flexible solution. It avoids complicated and potentially cost-intensive licensing issues. In addition, the virtualization platform is developed as a cloud service (Baun et al., 2011), operated and automatically set up by the university itself on-premise, running in the university's server infrastructure, but could, in principle, also be implemented as a public cloud service. The core element is the virtualization platform Proxmox (Algarni et al., 2018; Cheng, 2014). It is based on the Debian Linux operating system in conjunction with the KVM/QEMU (Kernel-based Virtual Machine / Quick Emulator) hypervisor solution. As Proxmox supports operation on several nodes of a cluster and comes with extensive control tools, it is ideally suited as a basis for operating virtual machines in a professional environment and for developing a virtual learning platform (Chen et al., 2017). The platform uses the VXLAN (Virtual Extensible Local Area Network) protocol (Mahalingam et al., 2014) to encapsulate the virtual machines in a separate network enclosure. Virtual networks allow the creation, modification, and removal of networks purely by software. It enables largely isolated operation of the individual users' virtual machines via independent IP networks. Figure 1 presents the architecture and software stack of the SKILL/VL platform.

The web interface (see Figure 2) allows users to access virtual teaching tools easily. Users can effortlessly set up, manage, and operate complex scenarios with virtual machines and networks, regardless of their technical expertise. Components can be selected via drag-and-drop, allowing users to create intricate networks effortlessly. Additionally, a simple click can add virtual hard disks and network cards. The interface includes a Creator for creating virtual learning rooms with templates for users to share. By executing a template, VMs and networks are automatically configured. This user-friendly interface caters to people from all backgrounds, irrespective of their computer science knowledge, built on NextJS and using the ReactFlow framework (Thakkar, 2020).

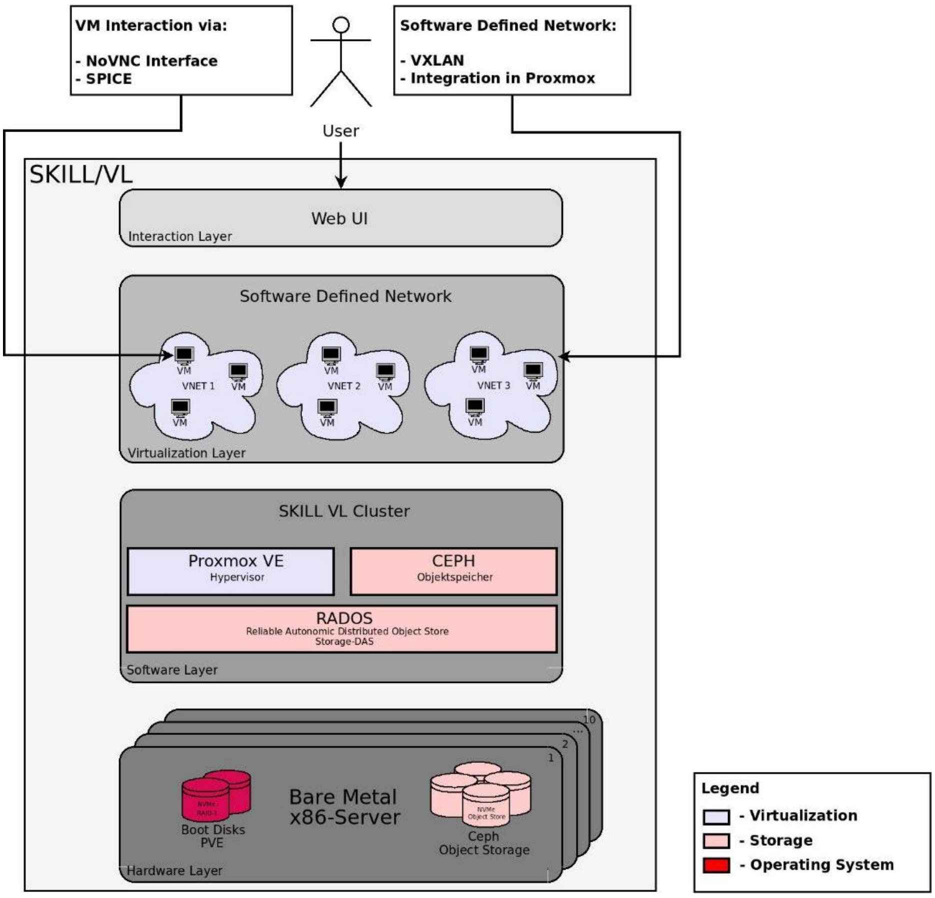

Figure 1. Architecture and software components

The interface offers students, academics, and teaching staff easy access to the entire system and the opportunity to quickly develop and use complex IT scenarios. It simplifies the design of new courses and simultaneously modernizes existing modules. Virtual resources allow students to work on practical teaching content from anywhere. There is currently no other free system with comparable features. The architectural design of the technical infrastructure is a hyperconvergent infrastructure. Hyper-converged IT architectures are an IT infrastructure in which all IT components required to operate the infrastructure are available centrally, and software-defined components are used to map them to hardware (Meneses et al., 2021). Figure 1 shows the details of the infrastructure.

Figure 2 presents the web interface of the SKILL/VL platform. Figure 2a shows the landing page of the platform with tiles offering information on the platform and the application of the platform in different use cases for students and lecturers alike. Figure 2b shows the graphical virtual room creator for the design of networking scenarios. It offers drag-and-drop functionalities for connecting the virtual machines with switches over graphical links resembling cables. This lowers the level of abstraction for users. Figure 2c presents the template selector, which offers to select configured and saved virtual rooms. The templates can be edited and started with different scenarios. Figure 2d shows the detailed permissions view used by lecturers for the configuration of access rights of virtual resources. This view is based on the component Keycloak (more details in section 5).

The objectives of the developed platform are the strict and consequential use of open-source software for its construction and operation and the development of an easy-to-use virtual environment for the practical enhancement of computer science classes. Section 5 presents the implementation of the components.

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Figure 2a. Landing page of SKILL/VL

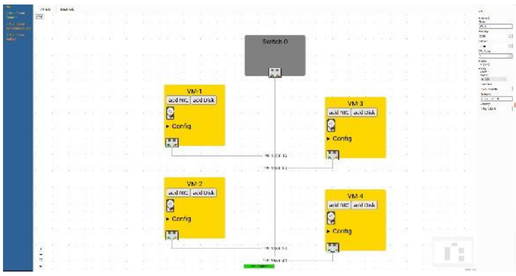

Figure 2b. Virtual room creator

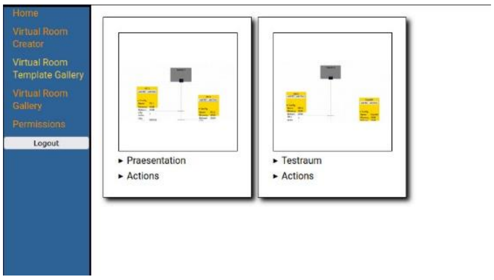

Figure 2c. Template selector

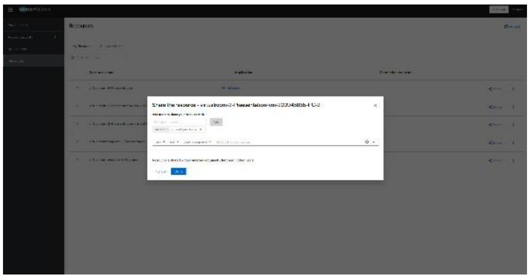

Figure 2d. Detailed permission view of resources

Figure 2. SKILL/VL user interface

5. IMPLEMENTATION OF INDIVIDUAL COMPONENTS

We prioritize using free software as a core philosophy, emphasizing established and reliable software frameworks. Figure 3 illustrates the infrastructure components and their interactions. An example is KeyCloak, which offers user authentication within the platform by linking infrastructure resources to the university's authentication service. KeyCloak enhances flexibility by enabling a centrally administered single sign-on service that can be expanded and connected to external authentication services.

As described, user interaction is realized via a web interface based on the NextJS framework. The templates for virtual learning spaces created here by the users can be shared and executed with other users. The sharing of resources is implemented via the Keycloak service using the UMA protocol (Maler et al., 2019). The storage and execution of virtual learning spaces (and templates) are implemented as a backend service developed in Python. This backend uses the FastAPI framework to communicate with the Proxmox hypervisor and implements a REST interface. This way, the virtual learning room's virtual machines and VXLANs are automatically created and configured. The use of VXLAN in the context of computer networking classes is beneficial, since it offers each user an individual address space, which is isolated from each other and on top of that it is hidden from the user. The networks used by the virtual learning environments are invisible to the end user and are configured automatically for the virtual room without the necessity of the user's interaction. Each virtual switch in a learning space represents its own VXLAN bridge, which means each learning space is encapsulated separately (see Figure 3).

The user can select the operating system installed from preconfigured (preseeded) ISOs in the web interface. The QEMU-Guest-Agent is already pre-installed with these operating systems. The backend uses QEMU-Guest-Agent to apply the user's configurations (e.g., static IP addresses). In addition, the backend (in the live operation of a learning room) enforces the users' permissions for the individual VMs. For example, users cannot access all VMs in a learning room via VNC. In this way, for example, a web server that students can only access via API can be part of a learning space. The backend is developed to be easily expandable and to ensure the possibility of using other interface technologies in the future. For example, the service could be interconnected with cloud service providers beyond the current data center with Amazon Web Services, Google Cloud Platform, or Microsoft Azure and expanded into a hybrid cloud or multi-cloud scenario.

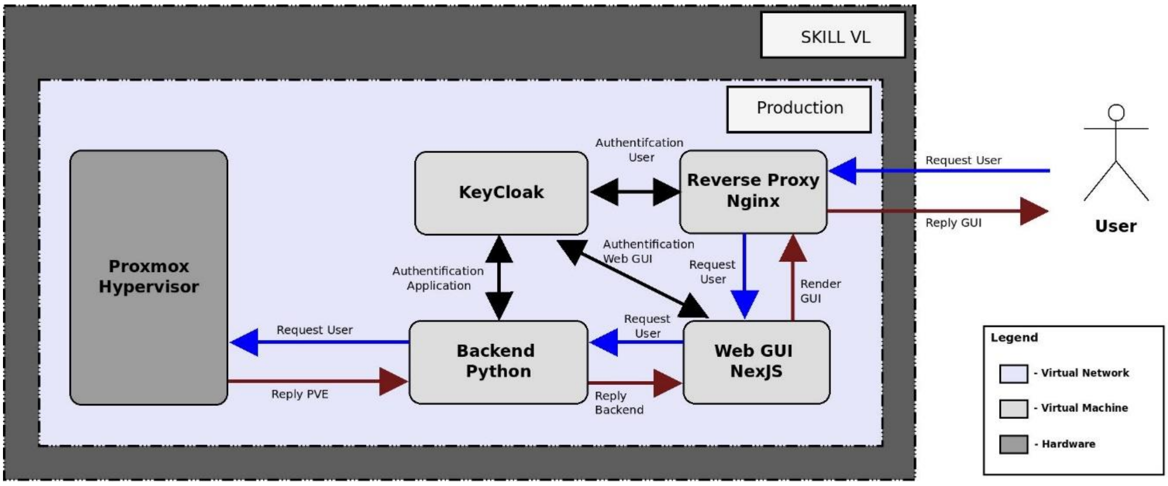

Figure 3. Software components of the SKILL/VL infrastructure and interaction

6. APPLICATION SCENARIOS

One example of an application for the platform that has already been evaluated in practice is the realization of practical exercises as part of a module on computer networks. However, the virtual environment's potential is broader than computer network courses. Other modules benefit from the use of virtual learning spaces. An application in the distributed applications courses allows the design of various distributed architectures. On the one hand, this enables students to visualize the architectures graphically and, on the other hand, to interact with the software components of the individual layers of the architecture. Layered architectures such as the three-layer architecture can be represented just as well as complex service-oriented architectures (Richards and Ford, 2021).

The SKILL/VL platform has a strong emphasis on extensibility, since it is designed for the use in different courses in computer sciences. This is not possible with other solutions like Mininet or GNS3, which capabilities are limited to computer networks. The next subsections demonstrate the use in different application scenarios.

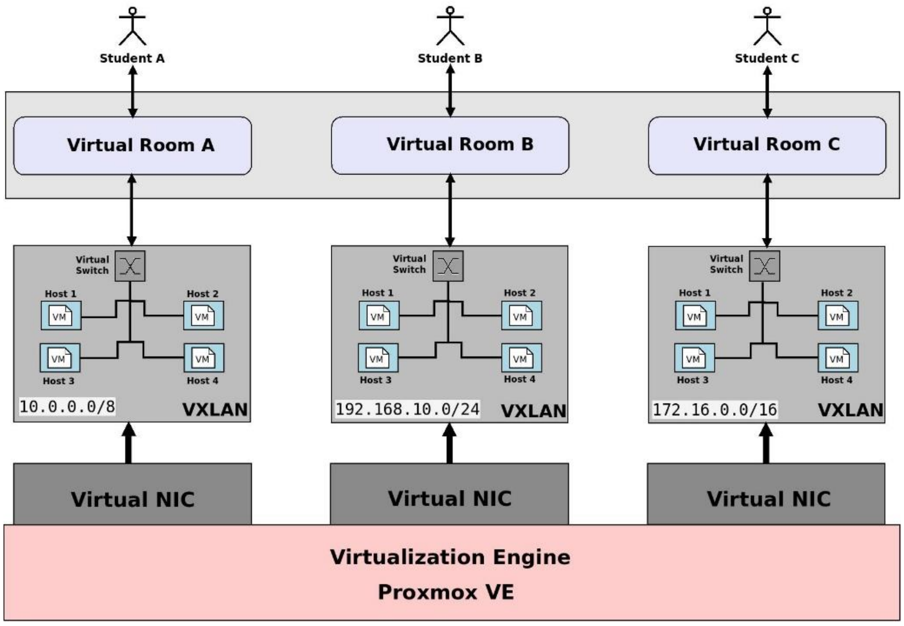

Figure 4. Mapping virtual learning spaces to physical components

6.1 Virtual Learning Space for Computer Networks

Figure 4 shows an example of implementing an exercise scenario from the perspective of students, teachers, and staff. These virtual rooms are physically separated through VXLANs (virtual networks). Such lab exercises are extensive and complicated, and the university provides a maximum of 24 computer workstations, which can be used in parallel in the physical lab for networks during normal university operations. This means that the capacity of the lab is minimal. In contrast, the infrastructure provides capacity for many students and can ease the burden on the university's operations. This technology is ideal for use as part of a practical networking course for setting up and experimenting with practical networking course for setting up and experimenting with large networks. It makes it easy to simulate an extensive corporate network at the touch of a button without much effort. Students can either work in teams or explore their corporate network individually. Investigating networks designed in is another practical use case for teaching. Such scenarios can be peppered with, for example, deliberate misconfigurations to train structured troubleshooting in realistic networks or teach security vulnerabilities. In addition, the configuration of security mechanisms, such as packet filters and firewalls, can be practically tested.

In the summer semester of 2023, an initial test run of the platform took place with computer science bachelor's students as part of the computer networks course. The users completed laboratory tasks, which were completed on physical hardware at the beginning of the semester, again in the project's virtual infrastructure. Specifically, a simple network consisting of four virtual machines with the Debian Linux operating system was configured, and the network traffic between the machines was analyzed using the Wireshark tool.

One problem for many students initially was the changeover from physical to virtual cabling, as this requires a greater capacity for abstraction. There were also initial challenges among the participants in understanding the web interface, as some of it was still experimental. Furthermore, when using the virtual machines, some students experienced delayed interactions (so-called lags) with the VNC client noVNC, which takes over the interaction via VNC protocol with the operating system instances on the virtual machines on the server side and converts the graphical output into a web stream for the browser on the user's end device.

6.2 Virtual Learning Space Internet-Of-Things

A further application example is developing a virtual Internet of Things (IoT) learning space. Physical sensors can communicate with virtual nodes in the platform via the network. The goal is to enable teachers and students to quickly build and test complex IoT applications virtually. With the framework, users can define the IoT topology and workflow in the web interface and then simulate the experiment in the virtual environment. Teachers, scientists, and students can use the web interface to build the IoT structure and visualize the construction activity during execution. It can give students a practical representation of cyber-physical systems (CPS) (Wolf, 2009). The logical flow of data from sensors to the cloud can be represented simply. Industry 4.0 scenarios can also be applied, and the complexity of the topic can be reduced by using a graphical editor (see Figure 2).

For example, virtual and real sensors like temperature and humidity and actuators like LEDs and controllers based on virtual Raspberry Pi single-board computers and Python scripts have been implemented. They interact in the virtual environment and communicate via Python sockets and the IP protocol with sensors and actuators in the virtual and real world. There are also plans to integrate other IoT protocols and communication layers, such as COAP and MQTT, and the integration of other transmission protocols, such as WiFi, Bluetooth, Zigbee, and Z-Wave.

7. CONCLUSION

The realization of a virtual learning platform for university teaching, which in particular allows creating and using complex IT structures for non-experts in teaching, was successfully realized using only free software. Students have successfully evaluated the developed web interface and the setup of the underlying infrastructure. The next step will be the productive use of the platform in various courses. Close monitoring through performance measurements and evaluations is planned to obtain data on the technical characteristics, such as the reliability and scalability of the service, as well as user suggestions for improvement.

8. FUTURE WORK

The project has shown the feasibility of its objectives so far. Further innovative ideas will be explored and implemented. Improvements to the web interface have been made to enhance user experience. An accessible version of the platform, following WCAG 2.1 guidelines, is needed for users with disabilities. Expanding the platform's capabilities includes offering various Linux distributions. While integrating Windows VMs is possible, the priority lies with open-source components. Refactoring software toward a container platform will boost efficiency and speed up release cycles using a cloud-native setup.

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